Security Evaluation of UWB: Secure Ranging and STS Integrity in IEEE 802.15.4z

1. Evolution of Access Security: From Signal Strength to Time-of-Flight:

The historical trajectory of access control has transitioned from the physical rigidity of mechanical locks to the digital abstraction of wireless protocols. While the past decades, saw punch cards replace keys and then later introduced RFID for contactless convenience, these legacy systems shared a common vulnerability: they operated without a verifiable "physical-layer truth." The reliance on Received Signal Strength Indicator (RSSI) in technologies like Bluetooth and early NFC created a fundamental security gap. An adversary can easily manipulate RSSI through signal amplification or "relay attacks," tricking a receiver into believing a device is proximal when it is actually meters away. Ultra-Wideband (UWB) addresses this through a strategic move to Time-of-Flight (ToF). By utilizing nanosecond-long pulses, UWB shifts the security boundary from the easily spoofed amplitude of a signal to the immutable laws of physics—measuring the exact time it takes for a signal to traverse a distance at the speed of light.

Comparative Analysis of Proximity and Access Technologies

Technology	Pros	Cons
Mechanical Locks	Simple; immune to electronic hacking.	Keys are easily lost/cloned; no digital audit trail or sharing.
Biometric Systems	Highly effective personal identification.	"Costly; the ""key"" cannot be shared among trusted parties."
Bluetooth / RSSI	Ubiquitous hardware support; low power.	Vulnerable to RSSI spoofing; inaccurate distance estimates.
NFC (Legacy)	Established standard; convenient for payments.	"Requires active ""taps"" (cm range); vulnerable to replay/relay."
UWB (ToF)	Centimeter-level accuracy; high relay-attack resistance.	Requires specialized hardware and cryptographic PHY protection.

ToF provides the necessary foundation for spatial awareness, but the physical layer requires specific cryptographic enhancements to prevent waveform manipulation. These protections are formalized in the IEEE 802.15.4z standard through the **Scrambled Timestamp Sequence (STS).**

2. Technical Architecture of the Scrambled Timestamp Sequence (STS)

The IEEE 802.15.4z amendment introduces the **High Rate Pulse (HRP)** mode to secure ranging measurements. Unlike previous standards, 802.15.4z does not rely on predictable preamble patterns for timestamping. Instead, it utilizes the STS—a cryptographically secured, bipolar sequence designed to ensure ranging waveforms remain unpredictable to an adversary.

The STS is a dynamic, cryptographically generated waveform designed to prevent attackers from predicting or faking the signal.
• **AES-128 Generation:** The STS consists of a sequence of 4,096 pulses generated using AES-128 encryption in counter mode (CTR).
• **Pseudo-Randomness:** Because the pulses are generated using a secret seed (shared during pairing) and a counter that increments with every transmission, the sequence looks like random noise to an attacker.
• **Verification:** The car and the phone calculate the expected sequence independently. When the signal arrives, the car checks if the received pulse pattern matches its calculated "template." If an attacker tries to guess the sequence or replay an old one, the pattern won't match, and the car will reject the measurement.

STS Packet Architecture (Config 3)

The "STS Packet Config 3" is the preferred architecture for secure ranging, placing the cryptographic sequence immediately before the **Ranging Marker (RMARKER**).

Component	Role in the Ranging Process
SYNC	Initial acquisition and timing/frequency sync using ternary preamble codes with ideal periodic autocorrelation.
SFD	Start-of-Frame Delimiter; signals the transition from synchronization to the PHY header.
PHR	PHY Header; conveys frame length and signaling parameters.
STS	The cryptographic core; a bipolar sequence providing the "physical-layer truth" for ranging.
RMARKER	The absolute reference time used for all reported distance measurements.

Mechanics of STS Generation and Spreading

The STS is generated via an AES-128 engine operating in counter mode, producing a pseudo-random bitstream using CTR mode of operation.

3. Threat Analysis: Potential Attack Vectors:

In high-stakes automotive and industrial environments, UWB must maintain the "distance-bounding" principle: the system must ensure that the reported distance is always an upper bound of the true physical distance.

UWB Specification Mapping and Availability

Layer	Specification Source	Availability Type	Relevant Attack Vectors
Physical (PHY)	**IEEE 802.15.4a/z/ab**	**Open/Published:** Standardised by IEEE for pulse shapes and ranging.	Ghost Peak, Cicada, ED/LC, Relay attacks, and Channel Injection.

Medium Access (MAC)	**FiRa Core / IEEE MAC**	**Member-Only (FiRa):** FiRa Core specs are exclusively for members. IEEE MAC is open.	Clock Offset (Mix-Down/Stretch-and-Advance), Spoofing, and Jamming (DoS).
Link / Transport	**FiRa / JUMPWG**	**Proprietary/Member-Only:** Defined by FiRa and the CCC-FiRa joint group.	Ranging Sequence Fragments (RSF) manipulation and synchronization errors.
Application	**CCC Digital Key**	**Standardized Framework:** Broad architecture is public; deep implementation and certification are member-based.	Memorization, Stalking, and application-level relaying

3.1 Attack Vector Origins by Layer:

• **Physical Layer (PHY):** Attacks here exploit the fundamental physics of radio propagation. **Relay attacks** target signal timing (ToF). **Ghost Peak** and **Cicada** attacks specifically target the "back-search" and leading-edge detection algorithms used by receivers to identify the first legitimate path of a signal. **Channel Injection** uses physical reflectors to introduce predictable multipath patterns to influence key generation.

• **MAC/Link Layer:** These vectors exploit how devices coordinate communication. **Clock Offset attacks** (like Mix-Down) manipulate transceiver imperfections to reduce measured distance. **Jamming** targets the SYNC field or preamble to prevent legitimate access.

• **Application Layer:** Vectors at this level focus on the security of the digital credential itself. **Stalking** and **Memorization attacks** rely on observing channel correlation over time or following a user to record signal characteristics for future unauthorized use.

UWB Communication Sequence & Threat Vectors:

Sequence Step	Description & Action	Components Involved	Potential Threat Vectors
1. Discovery & Negotiation	The vehicle broadcasts a beacon. The phone connects via **Bluetooth Low Energy (BLE)** to authenticate and exchange a **UWB Ranging Session Key (URSK)** derived from the master key.	• BLE Radio Secure Element (SE) App Layer	• **BLE Relay:** Relaying the wake-up signal to trigger the car remotely. **MITM:** Intercepting pairing parameters (mitigated by SE).
2. Synchronization	The transmitter sends the **SYNC (Preamble)** and	• UWB PHY (Preamble)	• **Cicada Attack:** Continuously injecting pulses to overwhelm the preamble and manipulate power estimation. **ED/LC:** Early

	Start of Frame Delimiter (SFD). The receiver uses this to detect the packet, synchronize the clock, and estimate the channel.	Analog Front End	Detection / Late Commit attacks targeting the preamble symbol timing.
3. Secure Ranging	The transmitter sends the **Scrambled Timestamp Sequence (STS)**, a pseudo-random pulse sequence generated via AES-128. The receiver correlates this against a known template to determine the precise Time-of-Arrival (ToA).	• UWB PHY (STS Field) AES-128 Engine	• **Ghost Peak:** Injecting random pulses to trick the receiver's back-search algorithm into locking onto a fake, earlier path. https://securepositioning.ch/ghost-peak/ https://www.usenix.org/system/files/sec22fall_leu.pdf https://www.usenix.org/system/files/sec22_slides-leu.pdf **Distance Reduction:** Manipulating the analog signal to shorten the measured ToF.
4. Data Transmission	If commands (e.g., "Unlock") are sent, they are carried in the **PHY Payload**. This data is encrypted at the application layer using the session keys established in Step 1.	• UWB PHY (Data Field) MAC Layer	• **DoS:** Broadcasting noise to corrupt the payload checksums, preventing command execution. **Eavesdropping:** Attempting to capture payload data (mitigated by encryption).
5. Measurement & Calculation	Devices exchange timestamps (Poll/Response/Final) to calculate the round-trip time and **Time-of-Flight (ToF)**. The system checks if the calculated distance is within	• MAC Layer (Timestamping) Local Clocks	• **Mix-Down / Clock Offset:** Manipulating the carrier frequency to exploit clock imperfections and reduce the calculated distance. **Stretch-and-Advance:** Stretching the signal in time to alter arrival timestamps.

	the allowed threshold.		

4. **Global Ecosystem Integration: CCC, FiRa, IEEE & CSA :**

The transition of UWB from a ranging tool to a mission-critical security protocol is driven by global standardization through the Car Connectivity Consortium (CCC), FiRa, and the Connectivity Standards Alliance (CSA).

- **CCC Digital Key 3.0/4.0:** This standard combines UWB for secure ranging with BLE for discovery and NFC for low battery "dead phone" access. It creates a robust, relay-attack-resistant entry system that is now standard across major automotive OEMs (Audi, BMW, Mercedes-Benz).
- **Industrial Verticals:** FiRa 4.0 leverages Uplink **Time Difference of Arrival (UL-TDoA)** for efficient tracking and hardware interoperability.
- **Smart Factories:** Real-time monitoring of assets and workers using low-power tags.
- **Public Transport:** Implementing Suspend **Ranging** to manage massive user density at gates without the latency of session re-establishment.

Conclusion:

The IEEE 802.15.4z standard and the Scrambled Timestamp Sequence have fundamentally redefined the trust model for wireless access. By anchoring security in the Reference STS Receiver's validation of Time-of-Flight, the industry has effectively neutralized distance-shortening and relay attacks, enabling a new era of secure, hands-free interaction.

Anticipated Outcome:

- Pick one of the attack vectors to showcase the proof of concept.
- Derive the new threat vectors in the UWB communication for any of the industrial use cases (Automotive/IoT/Operational Technology).

Appendix:

1. Physical and MAC Layer Foundations (IEEE)

The Institute of Electrical and Electronics Engineers (IEEE) defines the fundamental Physical (PHY) and Medium Access Control (MAC) layers for Ultra-Wideband (UWB) technology.

Year	Standard	Primary Security/Functional Features	Source Link
2007	**IEEE 802.15.4a**	Initial IR-UWB PHY; focused on precision indoor ranging and localization.	standards.ieee.org
2020	**IEEE 802.15.4z**	Introduced Scrambled Timestamp Sequence (STS) and High-Rate Pulse (HRP) mode for secure distance bounding.	standards.ieee.org
2025*	**IEEE 802.15.4ab**	(Draft) Adds Multi-Millisecond (MMS) operation and Narrowband Assistance (NBA) to extend range and reliability.	ieee802.org/15/pub/TG4ab

**Expected release date based on draft progress.*

2. Implementation and Hardware Interoperability (FiRa)

The FiRa Consortium focuses on the interoperability of UWB hardware and ensures that silicon and firmware from different vendors can communicate effectively.

Year	Specification	Key Advancements	Source Link
2021	**FiRa Core 1.0**	Initial framework for hands-free access control services.	firaconsortium.org
2023	**FiRa Core 2.0**	Expanded to location-based services and device-to-device communication.	firaconsortium.org
2024	**FiRa Core 3.0**	Added Hybrid UWB Scheduling (HUS) and the **CCC Digital Key UWB** feature for automotive use cases.	firaconsortium.org
2025	**FiRa Core 4.0**	Introduced Uplink Time Difference of Arrival (UL-TDoA) and support for Aliro digital keys.	firaconsortium.org

3. Application Layer and Ecosystem Standards (CCC)

The Car Connectivity Consortium (CCC) acts as the application-level architect, defining how smartphones and vehicles interact securely using multiple technologies.

Year	Specification	Core Technology and Application Scope	Source Link
2020	**CCC Release 2.0**	Standardised Near-Field Communication (NFC) for "tap-to-unlock" access.	carconnectivity.org
2021	**CCC Release 3.0**	Integrated UWB and Bluetooth Low Energy (BLE) for hands-free, location-aware passive entry.	carconnectivity.org
2025	**CCC Release 4.0**	Focuses on cross-platform interoperability, multi-level key delegation, and server-based owner devices.	carconnectivity.org